



第十二章 羧酸及其衍生物

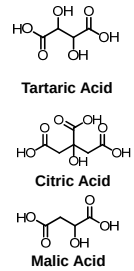
Carboxylic Acids and Carboxylic Acid Derivatives

引言

Found widely distributed in nature especially in foodstuff.

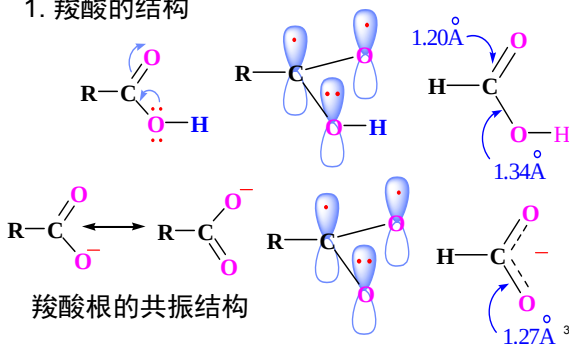


酸 Acids	来源 Source
酒石酸 Tartaric Acid	葡萄 Grapes
柠檬酸 Citric Acid	柠檬 Lemons
苹果酸 Malic Acid	苹果 Apples, 梨 Pears
乙酸 Ethanoic Acid	醋 Vinegar
甲酸 Methanoic (Formic) Acid	蚂蚁和蜜蜂的叮咬 Ant bites and bee stings
丁酸 Butanoic Acid	汗液 Sweat (body odour)



羧酸的分类、结构和命名

1. 羧酸的结构



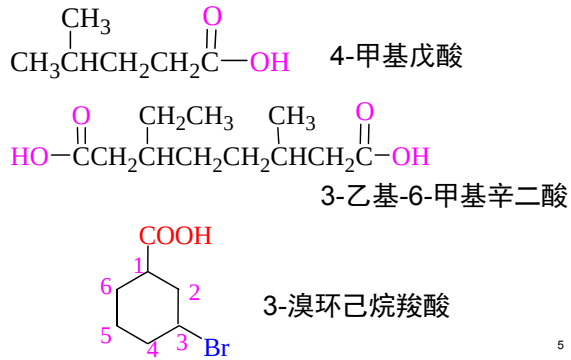
12.1 羧酸的结构、命名和物理性质

2. 羧酸的命名

系统命名

- 将相应的烃名改为 **某酸**，羧基始终编为1号
- 主链含烯键 → **某烯酸**；**某碳烯酸**
- 羧基直接连接在环上的：
 - 烃名 + 羧酸 —— **脂肪酸**
 - 烃名 + 甲酸 —— **芳香酸**

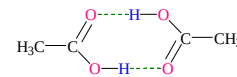
Example



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羧酸的物理性质

大多数羧酸在固态和液态时以二缔合体形式存在，bp比相应的醇高



乙酸的二缔合体

Hydrogen Bonding Impact on Boiling Point

Structure	Molecular Weight	Boiling Point
	58	0
	60	11
	64	12
	60	32
	58	49
	58	56
	60	82
	60	97
	60	118
	62	198

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1. 与水形成氢键

小分子的羧酸溶于水

2. 分子间形成氢键

熔沸点高，低级酸在蒸汽中 also 以二聚体形式存在

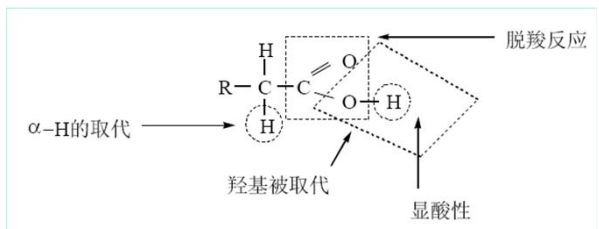
3. 气味

甲、乙、丙酸：强烈酸味与刺激性

C4—C9：腐败难闻 油状液态

正丁酸：动物汗液及奶油发酸变坏气味的起因

羧酸的化学性质



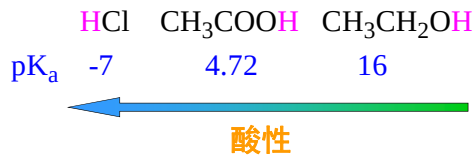
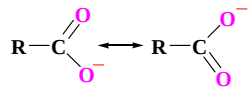
* 氯化态已最高，再加入氯，则表现为C-C断裂，即脱羧。

* 氯化态已最高，可降低氯化态，还原。

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酸性

- 酸性比醇强得多
- 仍是一种弱酸
- 一元饱和脂肪族羧酸的pKa值一般在3~5之间

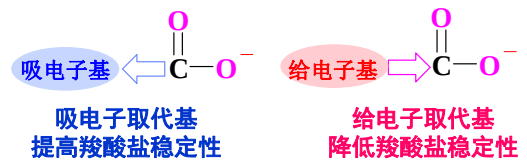


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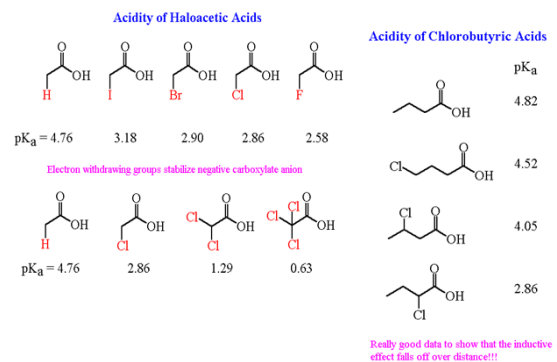
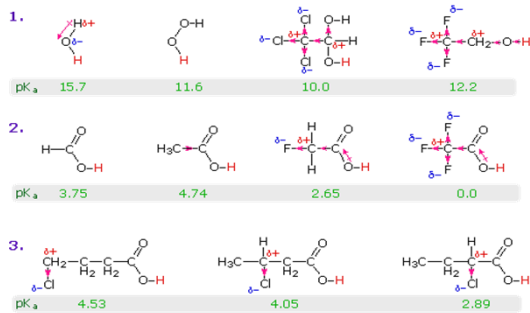
取代基对羧酸酸性的影响

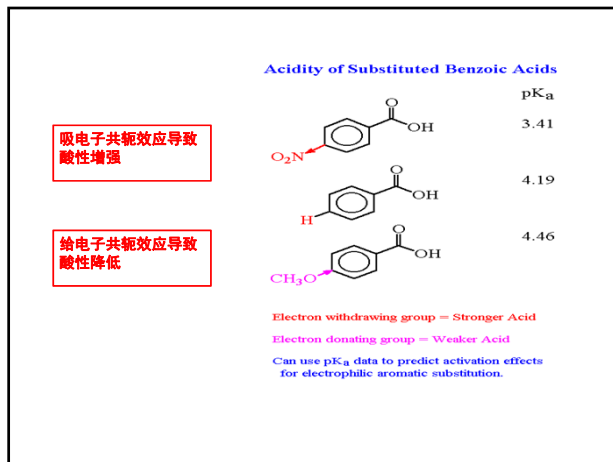
- 吸电子取代基使酸性增强
- 给电子取代基使酸性减弱

Structure	pKa
Cl_2CHCOOH	1.26
ClCH_2COOH	2.85
CH_3COOH	4.72



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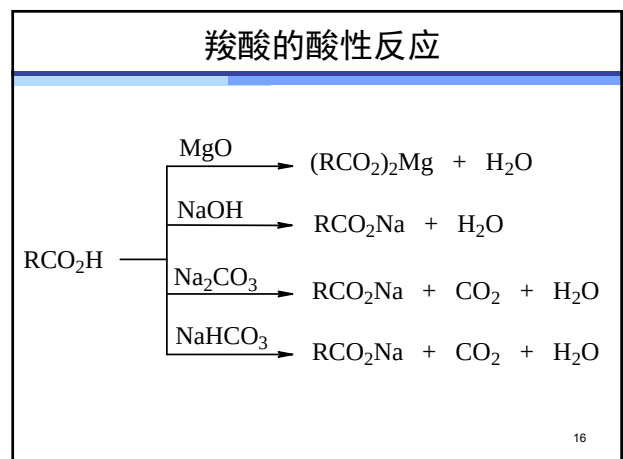
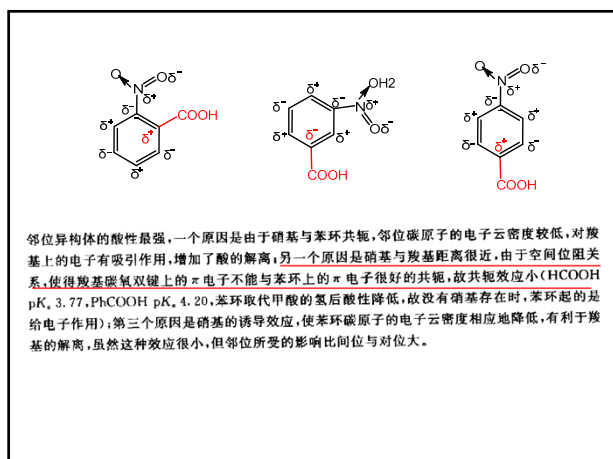




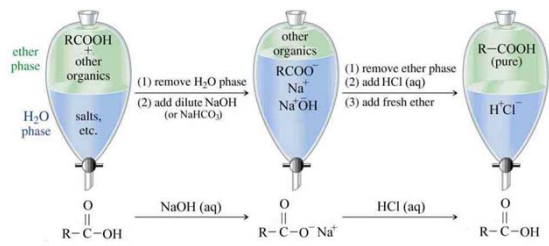
Acidity of Some Substituted Benzoic Acids

Y	<i>o</i>	<i>m</i>	<i>p</i>
—OH	2.98	4.08	4.57
—OCH ₃	4.09	4.09	4.47
—CH ₃	3.91	4.27	4.38
—H	4.20	4.20	4.20
—I	2.86	3.85	4.02
—Br	2.85	3.81	3.97
—Cl	2.92	3.83	3.97
—CN			3.55
—NO ₂	2.21	3.49	3.42

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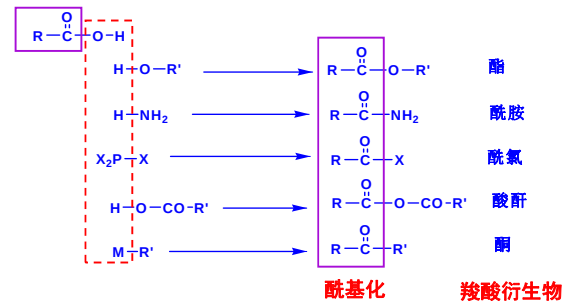


Purifying an Acid

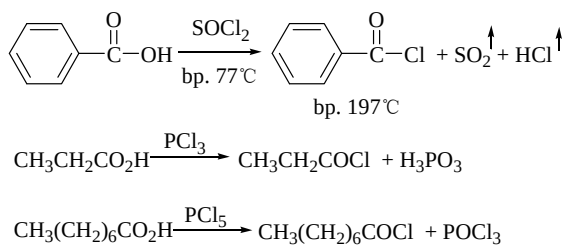


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2. 羧酸衍生物的生成

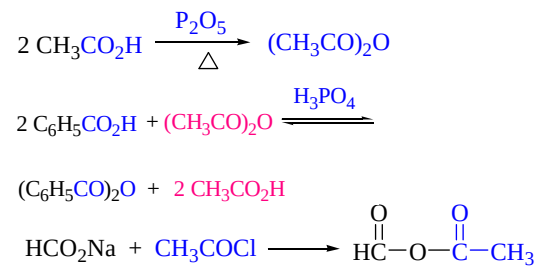


(1) 形成酰卤



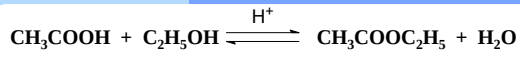
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(2) 形成酸酐

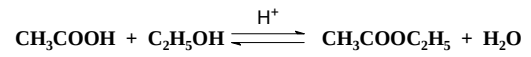
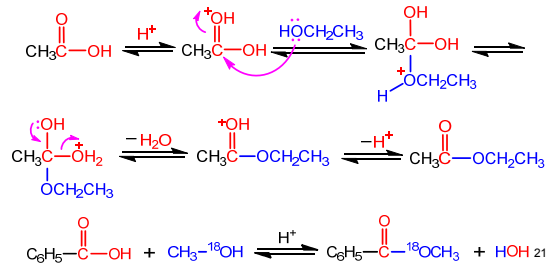


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(3) 生成酯



加成-消除机理



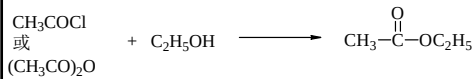
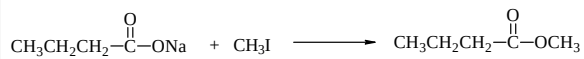
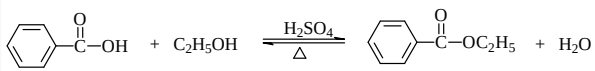
投料 1 : 1 产率 67%
1 : 10 97%

为了使正反应有利，通常采用的手段：

- ①使原料之一过量。
- ②不断移走产物。

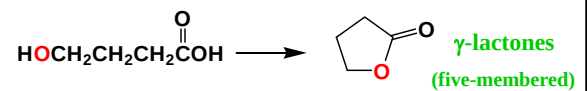
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(3) 生成酯



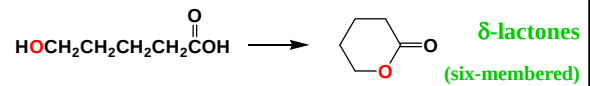
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分子内酯化--内酯 (Lactones)



4-Hydroxybutanoic acid

4-Butanolide

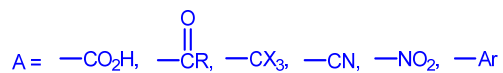


5-Hydroxypentanoic acid

5-Pentanolide

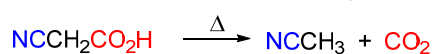
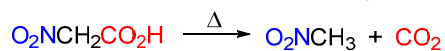
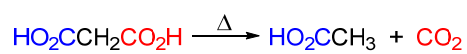
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脱羧反应



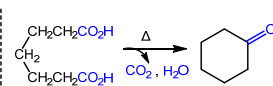
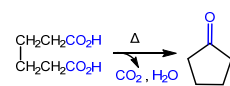
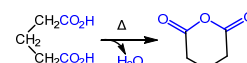
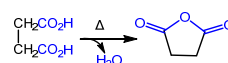
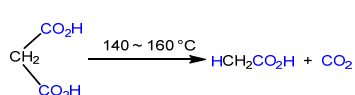
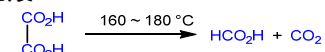
Electron-withdrawing groups

羧基的相邻碳上带有吸电子基团，在适当的条件下：碱、热、特殊催化剂，容易脱羧



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二羧酸脱羧



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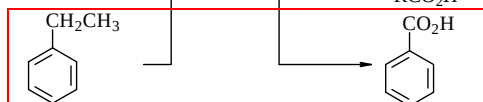
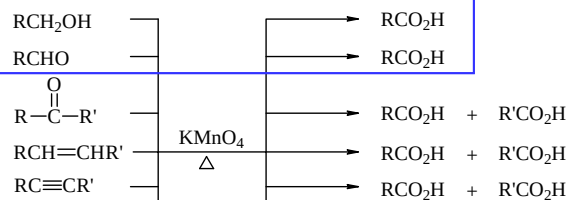
Blanc 规则

- 在有机反应中，如果有成环的可能时，一般总是形成五员和六员环。

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12.3 羧酸的制备

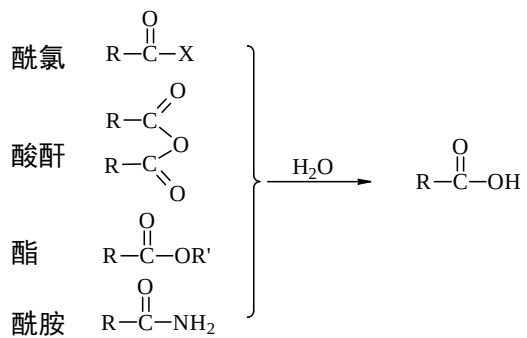
1. 通过氧化反应制备



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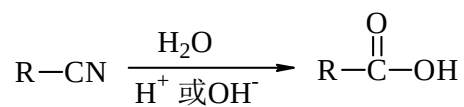
12.3 羧酸的制备

2. 由羧酸衍生物水解



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由腈水解

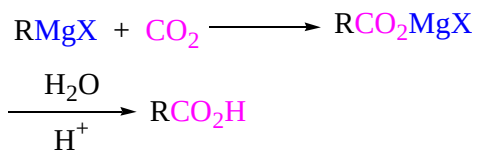


可由卤代烃制备增加一个碳原子的羧酸

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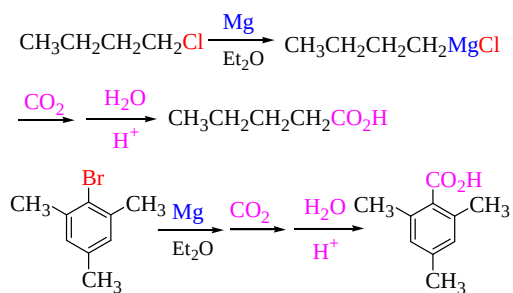
12.3 羧酸的制备

3. 格式试剂与二氧化碳反应制备



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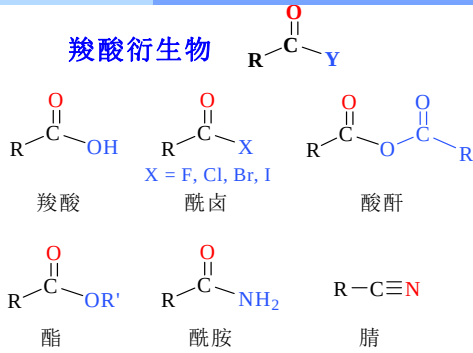
Example



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羧酸衍生物命名

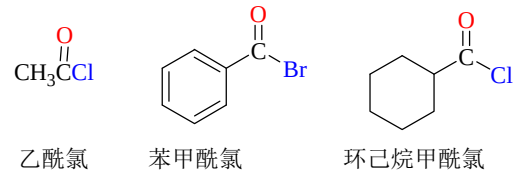
羧酸衍生物



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酰卤的命名: RCOX

酰基名 + 卤素名



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酸酐的命名: $\text{RCO}_2\text{COR}'$

× 酸 → × (酸) 酐; × 酸 × 酸酐



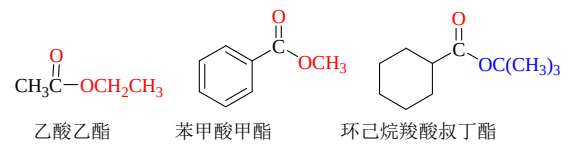
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酯的命名: $\text{RCO}_2\text{R}'$



羧酸的氢被烃基取代

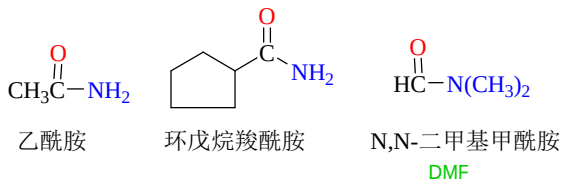
× 酸 × (烃基名) 酯



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酰胺的命名: RCONH_2

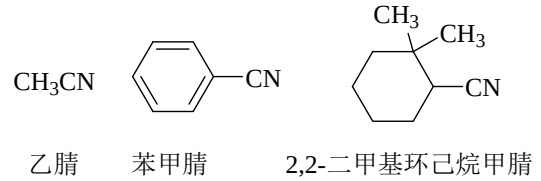
×酸 → ×酰胺 N-烃基×酰胺



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脩的命名: RCN

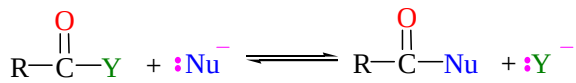
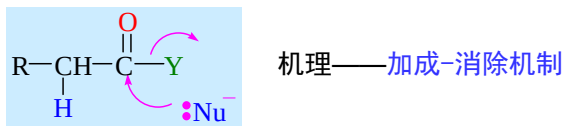
腈 × 腈(含CN碳原子)



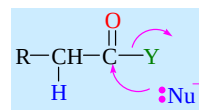
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羧酸衍生物的化学性质

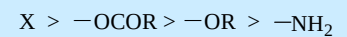
亲核取代反应



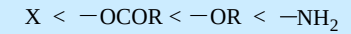
43



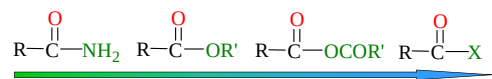
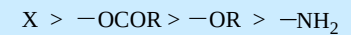
诱导效应



共轭效应



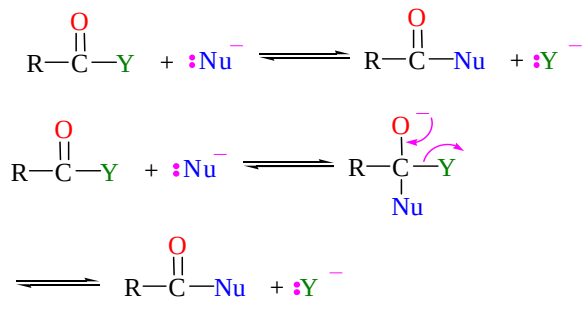
离去能力



反应性增强

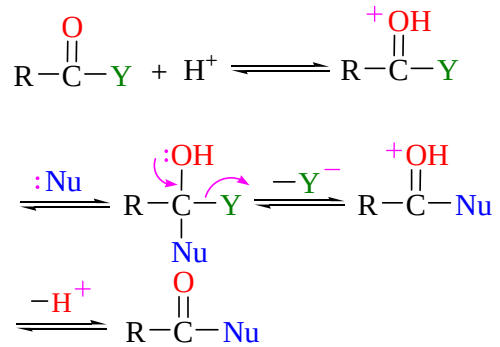
44

碱催化下的反应机理



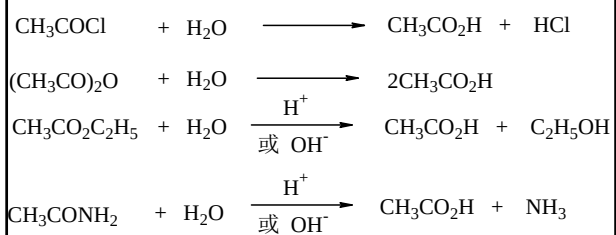
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酸催化下的反应机理



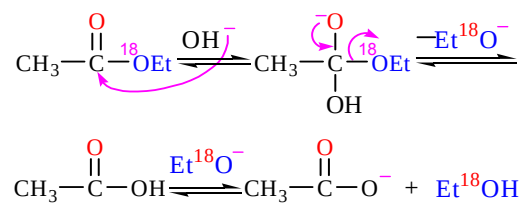
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1. 水解反应



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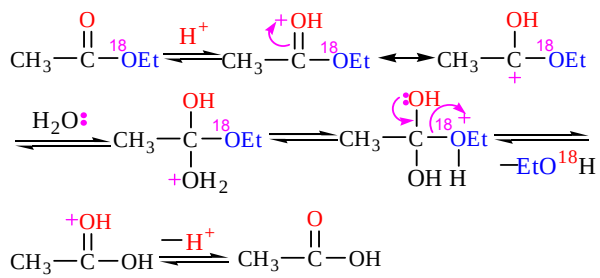
碱催化下酯的水解反应机理



经同位素证明酯的水解是酰氧断裂
而不是烷氧断裂

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酸催化下酯的水解反应机理

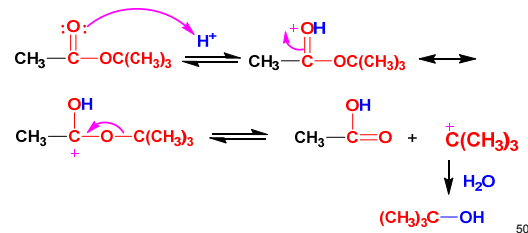


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3° 醇酯的水解

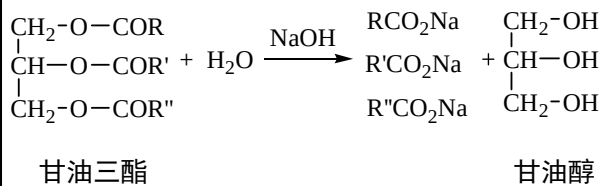
酸性条件下 $\text{CH}_3\text{CO}_2\text{R}$ 水解相对速率

R	CH_3	C_2H_5	$\text{CH}(\text{CH}_3)_2$	$\text{C}(\text{CH}_3)_3$
相对速率	1	0.97	0.053	1.15



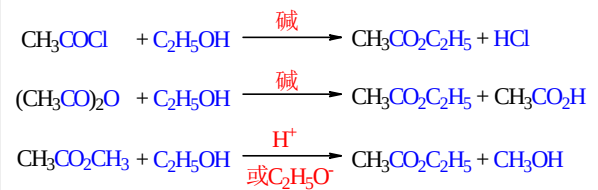
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天然油脂的皂化反应



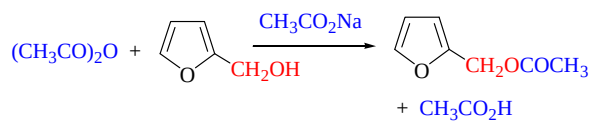
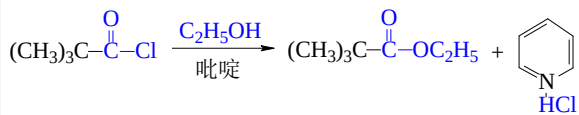
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2. 醇解反应



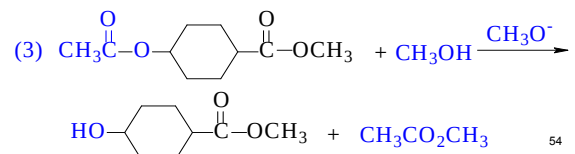
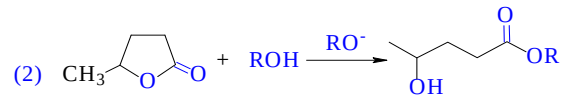
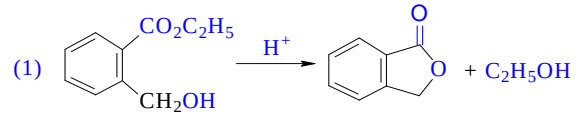
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醇解反应实例



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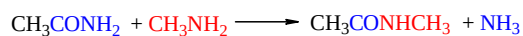
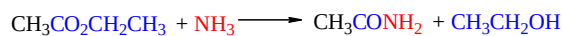
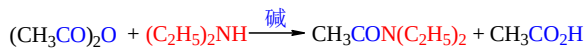
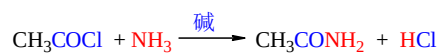
酯交换反应的实例



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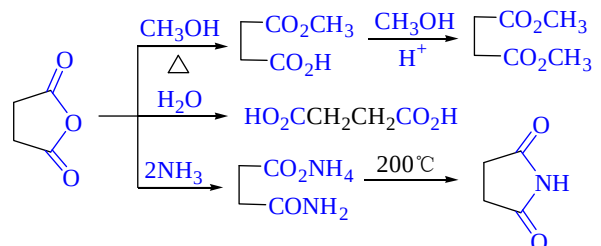
3. 氨解反应

- 羧酸衍生物氨解都得到酰胺
- 反应在碱性条件下进行较为有利
- 除3°胺外的胺均能与羧酸衍生物发生氨解反应



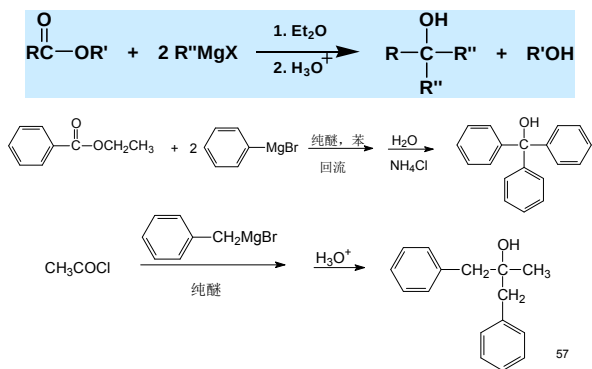
55

丁二酸酐的三解实例



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与格式试剂的反应

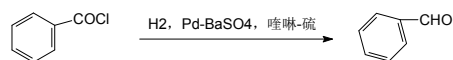


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羧酸衍生物的还原反应

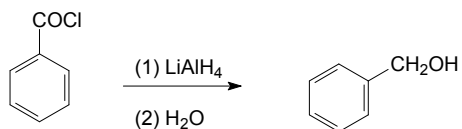
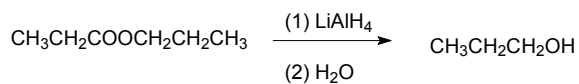
	RCOX	(RCO) ₂ O	RCO ₂ R'	RCONH ₂	RCN
H ₂ /cat.	RCH ₂ OH	RCH ₂ OH	RCH ₂ OH R'OH	RCH ₂ NH ₂	RCH ₂ NH ₂
B ₂ H ₆		RCH ₂ OH	slow	RCH ₂ NH ₂	RCH ₂ NH ₂
LiAlH ₄	RCH ₂ OH	RCH ₂ OH	RCH ₂ OH R'OH	RCH ₂ NH ₂	RCH ₂ NH ₂
NaBH ₄	RCH ₂ OH				

Rosenmund还原



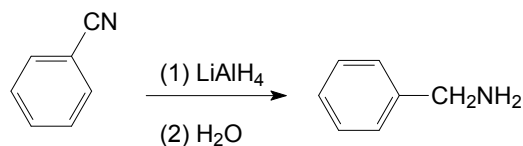
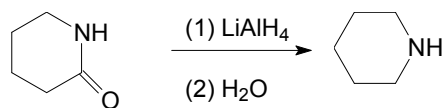
58

12.5.2 羧酸衍生物的还原反应

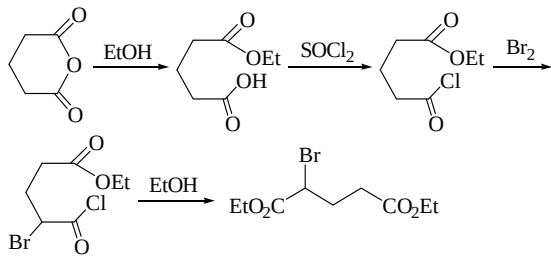
1. 用金属氢化物还原 (LiAlH₄)

59

Example



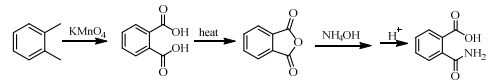
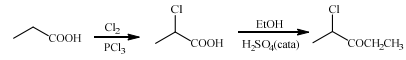
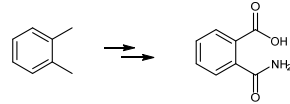
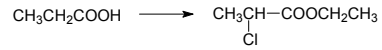
60

酰卤的 α -卤代

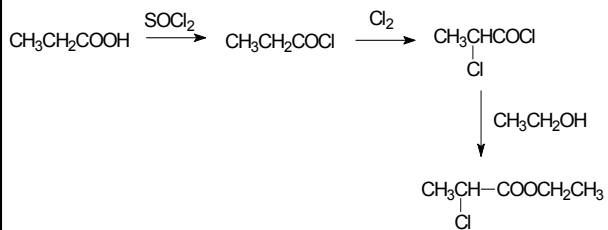
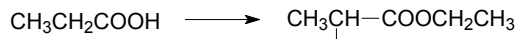
61

测试

请同学们完成下列转化

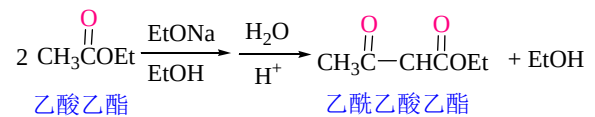


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酯缩合反应

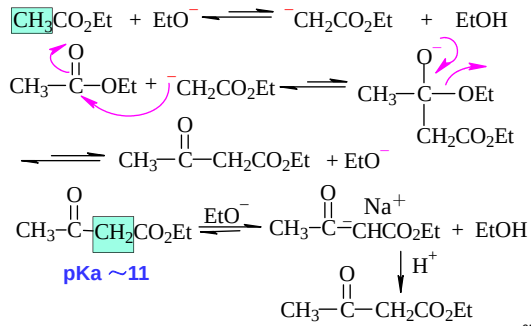
Claisen 酯缩合：含 α -氢的酯在碱的作用下，两分子酯发生缩合反应，生成 β -羧酸酯同时消去一分子醇



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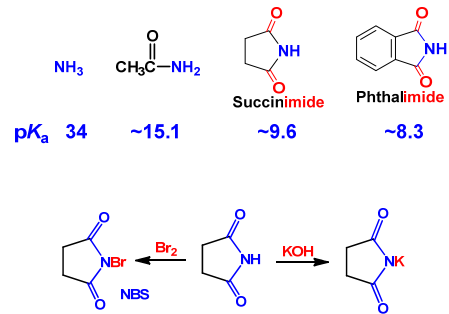
反应机理

pKa 24.5



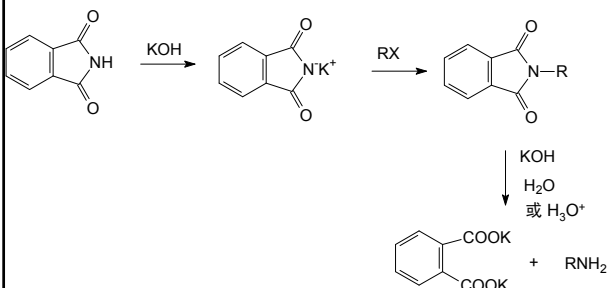
65

酰亚胺的性质



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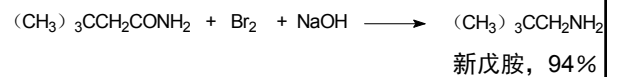
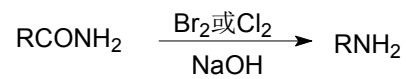
Gabriel反应----用于伯胺的制备



67

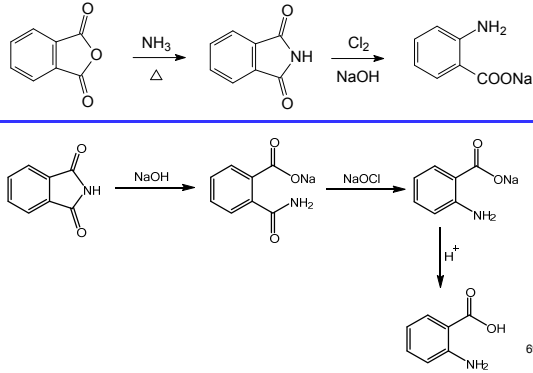
Hofmann降解反应

酰胺与溴或氯在碱溶液中作用，脱去羰基生成伯胺，使碳链减少一个碳原子的反应，通常称为Hofmann降解反应



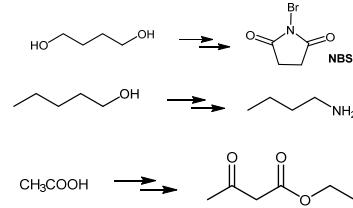
68

Example



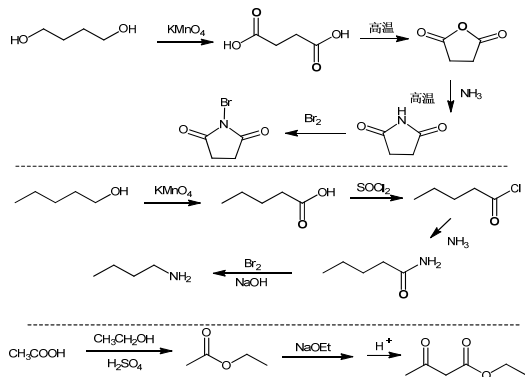
测试

请同学们完成下列转化



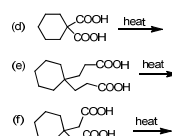
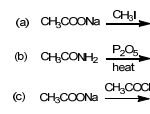
70

测试答案

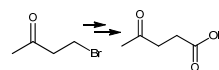


作业 (一)

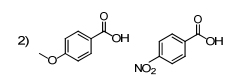
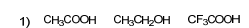
1. 完成下列反应



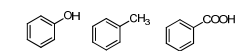
2. 完成下列转换



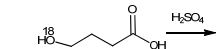
3. 排序题 按酸性由强到弱排序



4. 下列三个物质等重量的混在了一起, 设计实验方案分离纯化每一个化合物



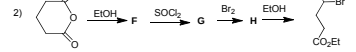
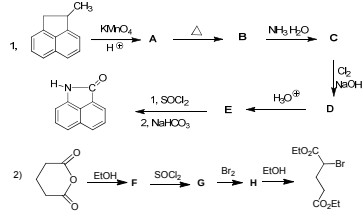
5. 写出下列反应产物以及反应机理



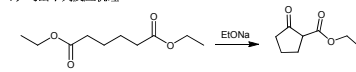
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6. 完成下列反应

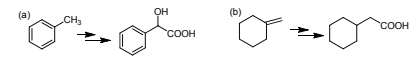
作业 (二)



7. 写出下列反应机理



8. 完成下列转换



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